

GNN-Based BERT for Understanding Context from Music

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Abstract—Understanding musical context requires jointly modeling acoustic structure and language about music. We present an end-to-end study that combines BERT-based text encoders with graph neural networks (GNNs) over music structure graphs for tagging, genre prediction, multimodal fusion, and caption–audio retrieval. Audio is represented with mel and chroma features; tracks are converted into chord-transition graphs and segment-similarity graphs. Four tasks are evaluated: (i) MusicCaps caption-to-tag classification with BERT, (ii) FMA-small genre classification with GraphSAGE versus a CNN mel baseline, (iii) GNN–BERT fusion for multi-label genre/tag prediction with DEAM valence–arousal auxiliary supervision, and (iv) dual-encoder contrastive retrieval on MusicCaps. On Task 1, BERT reaches macro-F1 of 0.657. On Task 2, GraphSAGE (accuracy 0.321) beats a majority baseline but underperforms a CNN mel model (0.485). Fusion ablations show that BERT-only remains strongest (macro-F1 0.110), while GNN-only and cross-attention do not improve tagging under our coarse 12-node chord graphs. Task 4 retrieval is near chance with limited paired audio ($R@10 \approx 0.03$), and a five-listener study yields a mean relevance of 2.62/5. The main takeaway is methodological: strong text cues and scarce audio–text pairs limit what simple structure graphs and late fusion currently buy, motivating richer graphs and larger paired corpora.

Index Terms—music information retrieval, BERT, graph neural networks, multimodal fusion, contrastive learning, music tagging

I. INTRODUCTION

Music is a multi-layered signal: a short clip may simultaneously convey genre and era, mood and emotion, harmonic motion, lyrical themes, and timbral texture. Modern music information retrieval (MIR) increasingly combines audio representation learning with language models [1], [2]. Pure sequence models on spectrograms capture local patterns well, but often underuse relational structure—how chord transitions relate, how segments repeat, or how those structures align with captions and tags.

Graph neural networks (GNNs) are a natural fit when music is cast as a graph of chords, segments, or events [3], [4]. BERT-style encoders [1] provide strong contextual text features from captions, tags, and metadata. The open question for student-scale multimodal MIR is not whether either modality works in isolation, but *when* graph structure and fusion help relative to strong text and CNN baselines under realistic data limits.

This paper studies that question through a unified pipeline with four complementary tasks:

- 1) **Caption tagging**: map MusicCaps captions to multi-label aspect tags using BERT.
- 2) **Structure-based genre classification**: classify FMA-small top genres from chord-transition graphs with GraphSAGE, compared with CNN mel and majority baselines.
- 3) **Multimodal fusion**: fuse graph and metadata/text encoders for multi-label genre and tag prediction, with DEAM valence–arousal as an auxiliary objective.
- 4) **Caption–audio retrieval**: train a dual encoder with InfoNCE for bidirectional retrieval and collect a small human relevance study.

Our contributions are threefold. (1) We release a reproducible feature-to-graph-to-model pipeline on FMA-small, MusicCaps, and DEAM, covering preprocessing, two graph schemas, training, metrics, and demo notebooks. (2) We run systematic ablations of GNN-only, BERT-only, early concatenation, and cross-attention fusion, plus majority and CNN mel baselines. (3) We provide an honest account of where coarse chord graphs and limited MusicCaps audio fail to beat simpler baselines—negative but informative results for hybrid MIR.

II. RELATED WORK

Music tagging and captioning. Large language-aware datasets such as MusicCaps [2] enable caption-conditioned tagging and text–music alignment. Transformer text encoders remain strong baselines for multi-label aspect prediction from captions. Classical tag datasets (e.g., MagnaTagATune) similarly motivate multi-label MIR evaluation with Macro-F1 and AUC-PR [5].

Audio deep learning for genre. CNN models on mel spectrograms are long-standing strong baselines for genre recognition [5]. Graph formulations of music—chord graphs, bar graphs, or similarity graphs—have been explored for structure-aware prediction [3], [4]. Our Task 2 tests whether a compact chord-transition GraphSAGE can compete with a mel CNN on FMA-small.

Emotion and continuous affect. DEAM provides continuous valence and arousal annotations [9]. We use DEAM as an auxiliary regression target in Task 3 rather than as a standalone emotion system, following multi-task formulations that couple tagging with affect.

Multimodal and contrastive MIR. Cross-modal retrieval with dual encoders and InfoNCE-style objectives has become standard for audio–text alignment [6], [7]. Fusion strategies range from early feature concatenation to attention across modalities. Our work does not claim state-of-the-art numbers; instead it measures how far a GraphSAGE + BERT stack goes on publicly available student-scale subsets, and where the stack breaks.

III. PROBLEM FORMULATION

Let a track be a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where X_{audio} denotes log-mel or chroma features over time, X_{text} is tokenized caption/metadata text, $G = (V, E)$ is a music structure graph, and y contains context labels (genres, tags, and optionally valence/arousal).

A BERT encoder maps text to contextual tokens $H_{\text{text}} \in \mathbb{R}^{L \times d}$ with CLS vector t . A GNN with L layers updates node features by neighborhood aggregation:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{AGG} \left(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right), \quad (1)$$

with $h_i^{(0)}$ initialized from chroma transition features or segment vectors. A graph readout $g = \text{MEANPOOL}(\{h_i^{(L)}\})$ is fused with text:

$$z = \text{Fusion}(g, t), \quad \hat{y} = \sigma(Wz + b). \quad (2)$$

For multi-label tagging we minimize binary cross-entropy, optionally plus an emotion auxiliary:

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \lambda(\alpha\|v - \hat{v}\|_2^2 + \beta\|a - \hat{a}\|_2^2). \quad (3)$$

Task 4 instead learns a shared embedding space with InfoNCE between normalized graph and caption vectors.

IV. METHOD

A. System Overview

Fig. 2 summarizes the pipeline conceptually: raw audio and text are preprocessed into features and tokens; chord or segment graphs are built; Task-specific heads then perform tagging (Task 1), genre classification (Task 2), fusion multi-label prediction (Task 3), or contrastive retrieval (Task 4). All stages are driven by a single `config.yaml` for sample rate, graph type, BERT name, GNN depth, and training hyperparameters.

B. Datasets and Preprocessing

FMA-small [8] is the primary audio corpus (8 balanced top genres: Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, Rock). We use splits of 6400/800/800 tracks (train/val/test) and retain about 7994 processed feature files. Audio is resampled to 22 050 Hz; we extract 128-bin mel spectrograms and 12-bin chroma, with 5 s segments and 2.5 s hop. FMA-medium is supported in code but not used in reported experiments due to compute and storage constraints.

MusicCaps [2] provides rich captions and aspect annotations (5521 public rows). Task 1 uses caption text and the top-50 aspect tags as a caption→tag proxy. Task 4 requires on-disk audio; we obtain about 1025 clips and

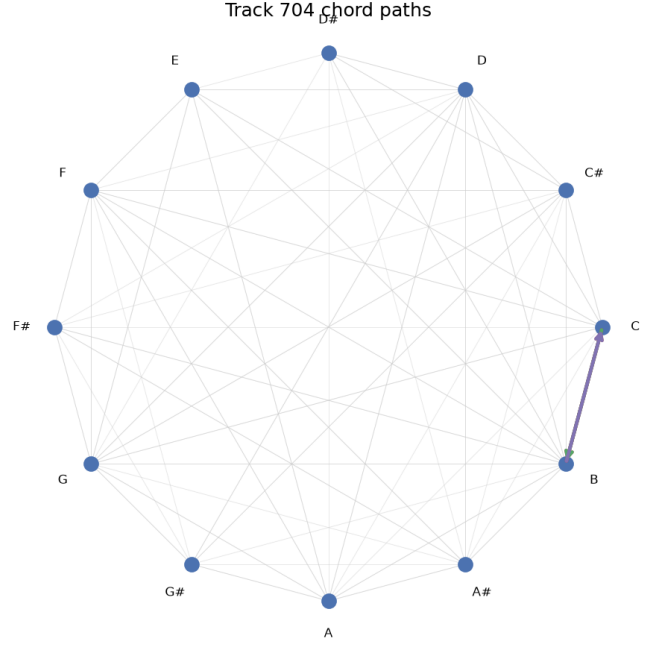


Fig. 1. Example chord-transition paths for FMA track 704 (International). Dominant transitions highlight recurring harmonic motion used as GNN input.

form 409/45/563 train/val/test pairs with available features—substantially smaller than the full caption catalog.

DEAM [9] supplies continuous valence and arousal annotations (1802 processed tracks; 1442/180/180 splits). It is used only as an auxiliary regression head in Task 3 ($\alpha = \beta = 0.5$, $\lambda_{\text{aux}} = 0.1$).

C. Music Structure Graphs

We build two graph families (Fig. 1).

Chord-transition graphs (Tasks 2–3). Each track yields a 12-node graph whose nodes are pitch classes. Edges encode observed chroma-argmax transitions with count weights; node features are outgoing transition histograms. This representation is compact and reproducible, but deliberately coarse: it collapses octave, inversion, and rhythm into pitch-class transitions.

Segment graphs (Task 4 / analysis samples). Nodes correspond to time segments with segment feature vectors. Edges combine temporal adjacency and cosine similarity above a threshold of 0.7, encouraging both sequential continuity and motif-like repetition.

D. Task-Specific Models

Task 1 (BERT tagging). We fine-tune `bert-base-uncased` with a linear multi-label head on the CLS token and binary cross-entropy over the top-50 MusicCaps aspects (max length 128):

$$t = \text{BERT}_{\text{CLS}}(X_{\text{text}}), \quad \hat{y}_k = \sigma(w_k^\top t + b_k). \quad (4)$$

Task 2 (GNN vs. CNN). A 2-layer GraphSAGE [3] (hidden size 128, dropout 0.2) mean-pools chord-graph node

embeddings for 8-way genre classification. The CNN mel baseline consumes the mel spectrogram without graphs or text. A majority-class baseline provides a floor for fair comparison.

Task 3 (fusion). A GraphSAGE encoder and BERT encoder produce graph and text embeddings from chord graphs and track metadata text (title/artist/album/tags). We ablate four modes: BERT-only, GNN-only, early concatenation of g and t , and cross-attention with queries from the graph and keys/values from BERT tokens:

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad z = \text{CONCAT}(g, AH_{\text{text}}). \quad (5)$$

The classifier predicts 164 multi-hot labels (114 FMA genres + 50 tags) plus valence/arousal heads.

Task 4 (contrastive retrieval). A dual encoder projects audio-graph and caption embeddings to a shared 256-D ℓ_2 -normalized space and is trained with InfoNCE [7] at temperature $\tau = 0.07$:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(g_i, t_j)/\tau)}. \quad (6)$$

We report Recall@{1, 5, 10} in both directions, zero-shot tag transfer diagnostics, and a small human rating study.

E. Training and Implementation Details

Unless noted, training uses AdamW, learning rate 2×10^{-5} , weight decay 0.01, batch size 16, and 20 epochs on a single CUDA GPU. BERT is fine-tuned (not frozen) in Tasks 1, 3, and 4. We checkpoint by validation Macro-F1 (Tasks 1/3), validation accuracy (Task 2), or validation retrieval (Task 4). Code is organized as `src/audio_features.py`, `graph_builder.py`, `bert_encoder.py`, `gnn_model.py`, `fusion_model.py`, `contrastive.py`, `train.py`, and `evaluate.py`, with notebooks for EDA, end-to-end demo, and Task 4 human scoring.

V. EXPERIMENTS AND RESULTS

A. Evaluation Protocol

For multi-label tagging we report Macro-F1, Micro-F1, and mean AUC-PR. For genre classification we report accuracy and Macro-F1. For DEAM auxiliary heads we report MAE and R^2 for valence and arousal. For retrieval we report Recall@ k in caption→audio and audio→caption directions. Human evaluation asks at least five listeners to rate whether the top-1 retrieved clip matches the query caption on a [1, 5] scale.

B. Task 1: Caption-to-Tag Classification

Table I summarizes test metrics. BERT reaches macro-F1 0.657, micro-F1 0.717, and AUC-PR 0.690, with validation macro-F1 peaking near 0.750. Fig. 3 shows steady validation F1 growth across epochs, indicating that captions alone encode many MusicCaps aspects well.

Qualitatively, high-confidence predictions often recover production and mood cues stated explicitly in the caption. For

TABLE I
TASK 1 TEST RESULTS (MUSICCAPS, TOP-50 TAGS).

Model	Macro-F1	Micro-F1	AUC-PR
BERT (caption)	0.657	0.717	0.690

TABLE II
TASK 2 TEST GENRE CLASSIFICATION ON FMA-SMALL.

Model	Accuracy	Macro-F1
Majority baseline	0.125	0.028
GraphSAGE (chord graph)	0.321	0.306
CNN mel	0.485	0.474

example, a low-quality ballad caption yields top tags *low quality* and *soulful*; an amateur finger-snapping vocal description yields *amateur recording*. Errors typically arise when captions imply instruments indirectly or use rare aspects underrepresented in the top-50 vocabulary.

C. Task 2: Genre Classification from Structure

Table II and Fig. 4 compare baselines on FMA-small. GraphSAGE (accuracy 0.321, macro-F1 0.306) substantially beats majority guessing (0.125) but trails the CNN mel model (0.485 / 0.474). Thus chord-transition graphs capture some genre signal—especially for styles with distinctive harmonic movement—yet dense spectrogram CNNs remain stronger when labels are the eight FMA top genres, which also depend heavily on timbre and production.

D. Task 3: Multimodal Fusion and Emotion Auxiliary

Table III reports the fusion ablation. BERT-only obtains the best test macro-F1 (0.110) and AUC-PR (0.252). Early concatenation is close (0.100) but does not improve over text. Cross-attention (0.044) and GNN-only (≈ 0.002) lag, suggesting that 12-node chord graphs add little complementary signal once metadata text is available, and that naive fusion can hurt optimization by injecting a weak modality into a strong one.

DEAM auxiliary MAE for valence/arousal is about 0.13/0.15 under BERT-only, but R^2 is negative, so emotion prediction is not reliable under this setup. The absolute Macro-F1 values are modest because the label space is large (164 multi-hot genre/tag dimensions) and many fine-grained genres are sparse on FMA-small.

Fig. 5 visualizes ablation Macro-F1. Figs. 6 and 7 show t-SNE projections of embeddings colored by genre and mood-oriented structure; clusters are partially separable by top genre but remain overlapping, consistent with imperfect multi-label discrimination.

Case studies on tracks 182 (Rock), 704 (International), and 890 (Folk) show interpretable dominant chord paths (e.g., recurring pitch-class oscillations), while predicted label rankings often reflect broad genre families more than precise fine tags. This supports the view that the graph encodes coarse

Pipeline overview (conceptual).

Raw audio / captions / metadata $\rightarrow$ Mel + chroma + tokenization $\rightarrow$ Chord / segment graphs $\rightarrow$ BERT / GraphSAGE / Fusion / Dual-encoder $\rightarrow$ Tags, genre, emotion heads, retrieval

Fig. 2. High-level system pipeline shared by Tasks 1–4. Task heads reuse the same feature and graph builders under one configuration file.

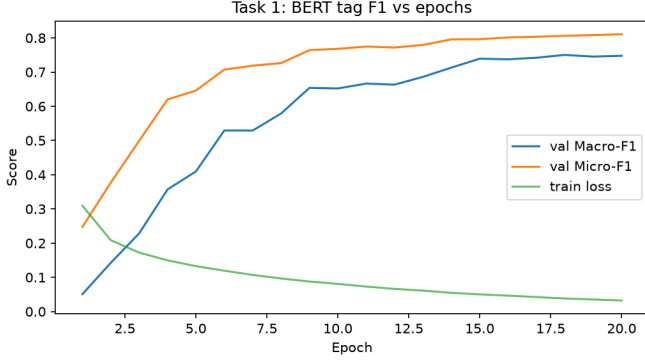


Fig. 3. Task 1 validation Macro-F1 / Micro-F1 and training loss over epochs.

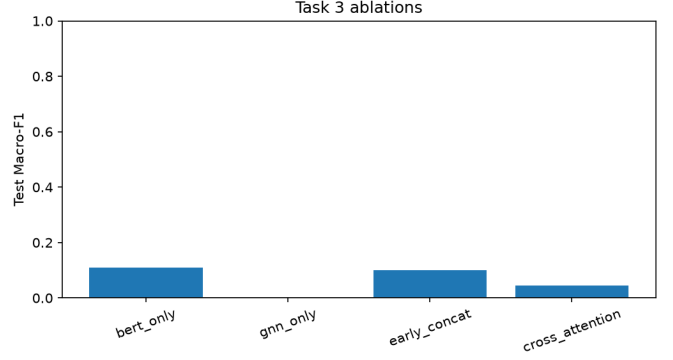


Fig. 5. Task 3 ablation: test Macro-F1 by fusion mode.

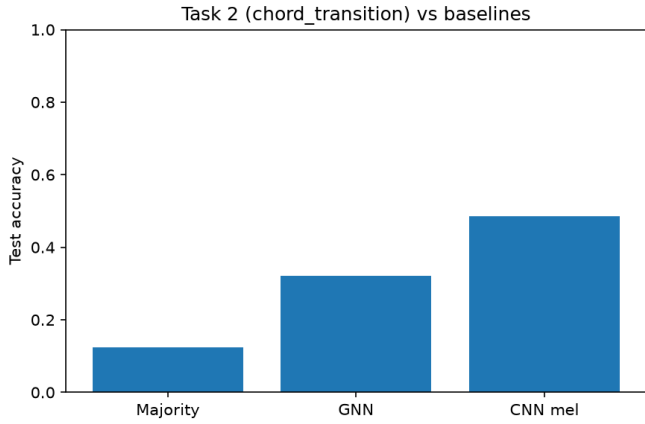


Fig. 4. Task 2 comparison of majority, GraphSAGE, and CNN mel baselines.

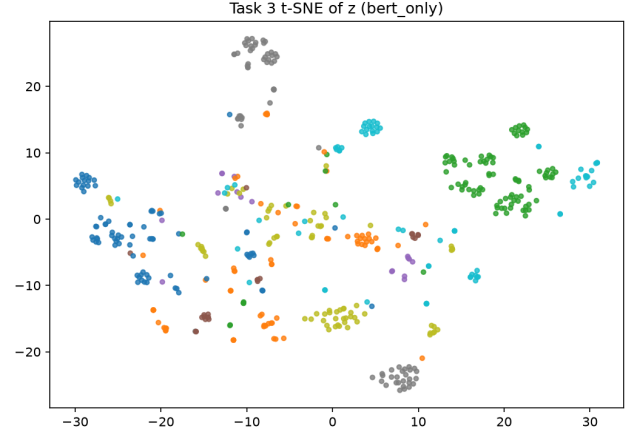


Fig. 6. t-SNE of Task 3 embeddings colored by FMA top genre.

TABLE III
TASK 3 TEST MULTI-LABEL METRICS (164 LABELS) WITH DEAM AUXILIARY.

Fusion mode	Macro-F1	Micro-F1	AUC-PR
BERT-only	0.110	0.258	0.252
Early concat	0.100	0.231	0.181
Cross-attention	0.044	0.223	0.109
GNN-only	0.002	0.018	0.031

harmonic habit, whereas metadata text carries most of the label-discriminative content.

E. Task 4: Contrastive Retrieval and Human Evaluation

With only 409 training pairs, test retrieval is weak (Table IV): caption $\rightarrow$ audio R@10 is 0.025 and audio $\rightarrow$ caption R@10 is 0.036 (Fig. 8). Qualitative top-3 lists frequently

mismatch the query semantics—for example, a sad piano-and-strings ballad query may retrieve flamenco guitar or bright folk textures with low cosine similarity (≈ 0.12 – 0.22).

Five listeners rated ten retrieval items on a 1–5 relevance scale (mean 2.62, std 1.40), consistent with limited top-1 correctness. Zero-shot tag prediction from contrastive caption embeddings is also weak (macro-F1 ≈ 0.104), far below Task 1 supervised caption tagging, confirming that the shared space did not learn robust semantic structure under our data budget. We treat Task 4 as a resource-constrained pilot rather than a competitive retrieval system.

F. Cross-Task Summary

Table V consolidates headline metrics across tasks. The strongest absolute result is Task 1 caption tagging. Structure helps versus chance in Task 2 but loses to CNN mel. Fusion

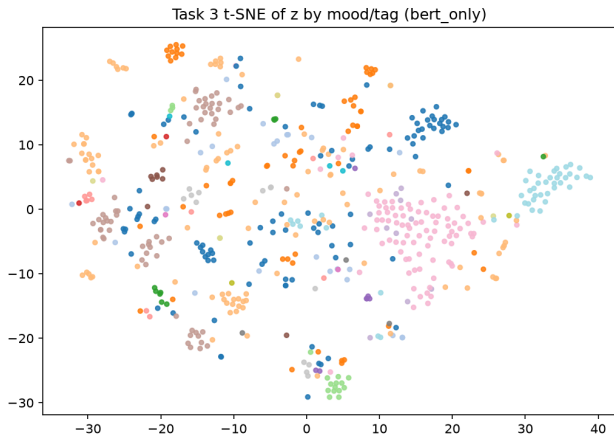


Fig. 7. t-SNE of Task 3 embeddings with mood-oriented coloring.

TABLE IV

TASK 4 TEST RECALL@ k ON MUSICCAPS PAIRS WITH AVAILABLE AUDIO.

Direction	R@1	R@5	R@10
Caption $\rightarrow$ audio	0.002	0.011	0.025
Audio $\rightarrow$ caption	0.002	0.018	0.036

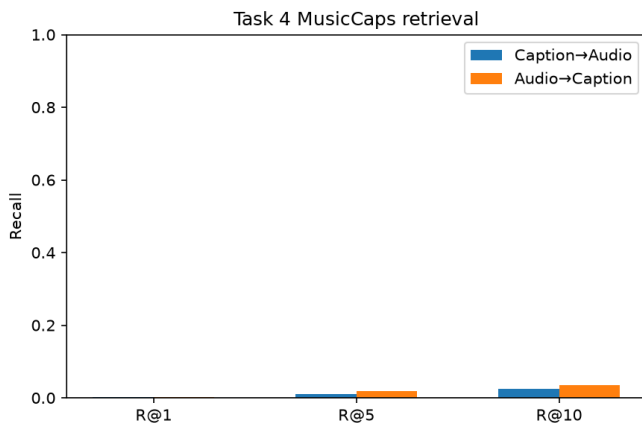


Fig. 8. Task 4 bidirectional retrieval Recall@ k on the test split.

TABLE V

HEADLINE EXPERIMENTAL SUMMARY ACROSS TASKS 1–4.

Setting	Key metric	Value	Note
Task 1 BERT tags	Macro-F1	0.657	Strong caption signal
Task 2 majority	Accuracy	0.125	Chance floor
Task 2 GraphSAGE	Accuracy	0.321	> majority
Task 2 CNN mel	Accuracy	0.485	Best genre model
Task 3 BERT-only	Macro-F1	0.110	Best fusion mode
Task 3 cross-attn	Macro-F1	0.044	Fusion regresses
Task 4 C $\rightarrow$ A	R@10	0.025	Data-limited
Task 4 human	Mean rating	2.62/5	5 listeners

does not beat BERT-only in Task 3. Retrieval remains near chance in Task 4 under limited paired audio.

VI. DISCUSSION AND FAILURE ANALYSIS

Several patterns emerge.

Language dominates when present. Task 1 succeeds with captions alone, and Task 3 prefers BERT-only over graph-heavy fusion. Metadata fields (title, artist, album, tags) already leak genre/tag information that a 12-node chord graph cannot match.

Structure helps versus chance, not versus spectrograms. Task 2 GraphSAGE beats majority but loses to CNN mel. Genre on FMA-small depends strongly on timbre and production; chroma transitions alone are an incomplete proxy for that signal.

Naive fusion can hurt. Cross-attention underperforms BERT-only, suggesting that injecting a weak modality without gating or residual text pathways can dilute discriminative features. Early concat is safer than cross-attention here, but still does not improve Macro-F1.

Contrastive learning needs scale. Incomplete MusicCaps audio yields near-chance R@ k and middling human scores. InfoNCE with batch size 16 and only hundreds of training pairs cannot form a stable global ranking over hundreds of test candidates.

Limitations. We use FMA-small rather than FMA-medium; chord graphs are coarse; Task 4 audio coverage is incomplete; human evaluation is small (5×10); and emotion R^2 is negative. These limits are deliberate trade-offs for a reproducible student project, but they bound the strength of positive claims.

VII. FUTURE DEVELOPMENT

We outline concrete next steps suggested by the failures above.

- 1) **Richer music graphs.** Replace 12-bin pitch-class nodes with beat-synchronous chord vocabularies, key-aware nodes, or hierarchical bar/phrase graphs; add rhythmic onset edges and instrument-conditioned subgraphs.
- 2) **Stronger audio encoders.** Initialize node or track features from pretrained audio models (e.g., spectrogram transformers) before GNN message passing, so graphs refine rather than replace timbral evidence.
- 3) **Better fusion.** Use gated residual fusion ($z = t + \text{Gate}(g)$), modality dropout, or late score fusion so text remains a default pathway when graphs are uninformative.
- 4) **Scale paired data.** Download a larger MusicCaps audio subset (or switch to datasets with guaranteed audio-caption pairs) and increase batch size / queue-based negatives for InfoNCE.
- 5) **Emotion transfer.** Train affect heads on DEAM with matched audio domains, or distill continuous valence/arousal into discrete mood tags before multi-task fusion.
- 6) **Broader evaluation.** Expand human studies, report per-genre confusion matrices, and compare against additional baselines (e.g., PCA+MLP on hand-crafted features).

- 7) **Interactive demo.** Extend `demo_context.ipynb` into a lightweight app that visualizes chord paths beside retrieved captions for error analysis.

VIII. REPRODUCIBILITY

Experiments are controlled by `config.yaml` (seed 42). Metrics are stored in `results/metrics.json`; plots in `results/plots/`; retrieval examples in `results/retrieval_examples/`; graph samples (≥ 20) in `results/graph_samples/`; and case studies in `results/case_studies/`. Large model checkpoints are kept locally and excluded from version control due to repository size limits; training commands recreate them from the public datasets and configuration.

IX. CONCLUSION

We presented a GNN-BERT pipeline for music context across tagging, genre classification, multimodal fusion, and retrieval. BERT caption tagging works well; chord graphs beat majority genre guessing but lose to CNN mel; fusion does not surpass BERT-only under our setting; and retrieval remains data-limited. Beyond the numbers, the study clarifies a practical design rule for hybrid MIR: add structure graphs and fusion only when they supply information that text and spectrograms do not already contain, and match model ambition to paired-data scale. These lessons, and the future directions above, provide a concrete path from a student-scale prototype toward stronger multimodal music understanding.

REFERENCES

- [1] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [2] A. Agostinelli *et al.*, “MusicLM: Generating music from text,” arXiv:2301.11325, 2023.
- [3] W. Hamilton, Z. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. NeurIPS*, 2017.
- [4] J. Jeong, S. Kim, and K. Lee, “Graph neural networks for music score analysis and generation,” in *Proc. ISMIR*, 2019.
- [5] K. Choi, G. Fazekas, M. Sandler, and K. Cho, “Convolutional recurrent neural networks for music classification,” in *Proc. ICASSP*, 2017, pp. 2392–2396.
- [6] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021.
- [7] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” arXiv:1807.03748, 2018.
- [8] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, “FMA: A dataset for music analysis,” in *Proc. ISMIR*, 2017.
- [9] A. Aljanaki, Y.-H. Yang, and M. Soleymani, “DEAM: MediaEval database for emotional analysis in music,” in *Proc. MediaEval*, 2015.
- [10] B. McFee *et al.*, “librosa: Audio and music signal analysis in Python,” in *Proc. SciPy*, 2015, pp. 18–24.
- [11] M. Fey and J. E. Lenssen, “Fast graph representation learning with PyTorch Geometric,” in *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.